FoveaWAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

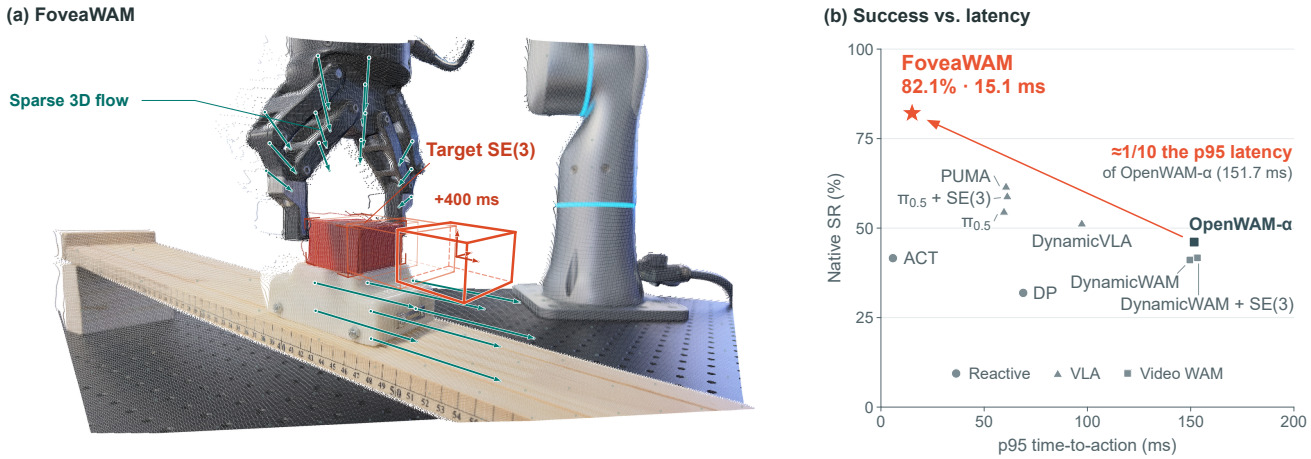


Figure 1. **FoveaWAM allocates prediction precision by task role.** (a) A precise SE(3) trajectory for the target and sparse 3D flow for its surroundings. (b) SR versus p95 time-to-action on OnTimeBench.

Abstract

To manipulate a moving object, a policy must predict the scene at the moment its action executes, yet every millisecond spent predicting pushes that moment further away. Existing policies either forecast the whole scene at uniform fidelity and act late, or act fast by ignoring everything but the target. Our key insight is that what a policy must predict decouples by role: only the target's pose, which decides contact, needs millimeter precision, while the surroundings matter only through the clearance they leave and the motion they impart. We present FoveaWAM, a native 4D world–action model that allocates precision by role: it models the target's past and future motion as a metric SE(3) trajectory and the surroundings as sparse 3D flow, and relies on the current frame for everything static. This compact representation lets FoveaWAM act in 15.1 ms on one RTX 5090, about $10\times$ *faster than video world–action baselines, giving the scene little time to move. To evaluate each policy under its own latency, we introduce OnTimeBench, a dynamic manipulation benchmark in which the scene keeps moving while the policy computes. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, and also leads on DOMINO and real-robot tasks.*

1. Introduction

To manipulate a moving object, a robot must act on a scene it has not yet seen. Objects keep moving while the policy perceives, computes, and acts, so every action lands in a later scene than the one observed. A ball rolling down a ramp at about 0.9 m/s moves 18 cm during a 200 ms inference; contact tolerates millimeters. The policy must therefore predict where the scene will be when its action executes. But prediction is itself computation, and its cost lengthens the interval it must cover: the longer a policy computes to anticipate motion, the further ahead it must anticipate. Existing dynamic policies settle this tension at one of two ends (Fig. 2). Some forecast the whole scene at uniform fidelity, as future video, latents, or 3D dynamics [31, 32, 49, 56, 61, 64]; they capture everything but act late. Others forecast only the manipulated object [14, 17]; they act sooner but miss the surroundings that block or redirect it. Either way, every part of the scene that is predicted costs the same, so knowing more means acting later. We ask instead: *what must a policy know about a moving scene, and how precisely, to make contact in time?*

Our key insight is that the answer decouples by role (Fig. 1). The target decides contact, so its pose at the moment of execution must be known to millimeter precision. The surroundings matter only through the clearance they leave

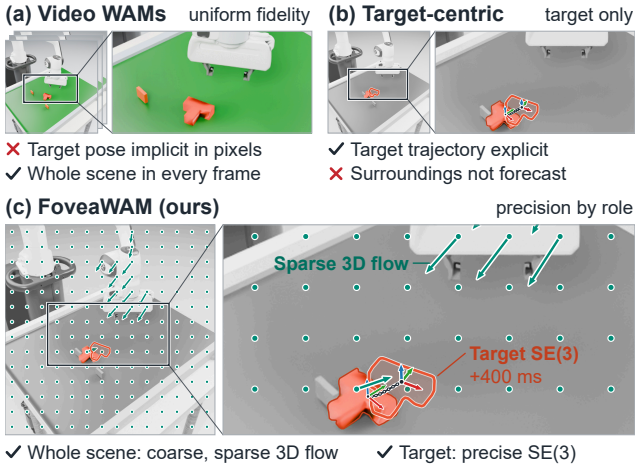


Figure 2. **Where forecast precision goes.** (a) Video WAMs forecast the whole scene and leave the target pose implicit [49, 56]. (b) Target-centric predictors forecast only the target's trajectory [14, 17]. (c) FoveaWAM forecasts a precise SE(3) trajectory for the target and sparse 3D flow for the scene.

and the motion they impart to the target, so their motion need only be known coarsely. Everything static is already in the current frame and needs no prediction at all. What must be known precisely is therefore one object's trajectory, not the scene, and a policy that spends its precision there can be exact where contact is decided and still act in time. Biological vision makes the same allocation: the fovea resolves fine detail only where gaze lands, the periphery stays coarse yet alert to motion, and gaze runs ahead of the hand to where contact will occur [12, 19, 24, 36, 42].

We build this allocation into **FoveaWAM**, a native 4D world–action model whose representation follows the three roles (Fig. 2c). The target is described by a metric SE(3) trajectory: a short pose history supplies its past motion in place of past image frames, and the model learns to forecast its continuation. The surroundings are described by sparse 3D flow, and everything static by the current frame alone. Both forecasts supervise the representation during training and are dropped at inference [60]. As a result, FoveaWAM acts in 15.1 ms on one RTX 5090, about 10× faster than video world–action models, leaving the scene little time to move.

Measuring this advantage requires an evaluation in which latency has a cost. The default evaluators of DOMINO and PhysMani-Bench pause the world while the policy computes [14, 61], so a slow policy is scored as if it were instant. We introduce **OnTimeBench**, 14 dynamic manipulation tasks in which the world never pauses: the scene keeps moving for as long as each policy computes, and every method is evaluated at its native latency. The change of protocol reorders existing policies: DynamicWAM with SE(3) history beats the lightweight ACT [66] by 23.4 points with the world paused, but by only 0.1 points at native latency. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline. Its lead holds with the world paused and on DOMINO [14], and it succeeds in 37.5% of trials across six real-robot tasks. Ablations support the allocation: coarsening the target's forecast or removing its pose history lowers success. Uniform fine flow recovers overall success but trails on both the rotation-sensitive and context-active task groups.

Our contributions are threefold:

- We show that what a dynamic policy must know decouples by role: millimeter precision for the target's pose at execution time, coarse motion for the surroundings, and no prediction for the static scene.
- We present **FoveaWAM**, a native 4D world–action model that allocates precision by role and acts in 15.1 ms, about 10× faster than video world–action models.
- We introduce **OnTimeBench**, a dynamic manipulation benchmark that evaluates each policy at its native latency and reorders existing policies relative to paused evaluation. FoveaWAM reaches 82.1% success on it, 20.7 points above the strongest baseline, and also leads on DOMINO.

2. Related Work

Dynamic manipulation. Acting on moving objects, long handled by engineered tracking and interception [1–3, 34, 40], is now learned from data [9, 26, 54]. Learned policies take motion from image-space cues, such as optical flow in PUMA [14] and DynamicWAM [31], or act on test-time forecasts of the scene [32, 44, 61, 64]. FoveaWAM instead reads the target's metric SE(3) pose history and observes only the current frame. Execution schemes [6, 29, 38, 45, 54] change when actions are applied, not what the policy represents, and are complementary. DOMINO [14] and PhysMani-Bench [61] provide diverse dynamic tasks but by default pause the simulation until the policy returns; OnTimeBench advances it during inference, so each policy is evaluated at its own latency.

Forecasting in world–action models. WAMs generate future video and actions in one network [25, 49, 56, 68], building on video prediction as a policy objective [46, 53]. Because video is costly to generate, recent work forecasts cheaper targets, such as features, flow, object slots, or compact latents [30, 43, 57, 59, 63, 67], and some use the forecast only in training [14, 60, 65]. In all of them the forecast lives in pixel, feature, or latent space, and the target's future pose stays implicit. FoveaWAM keeps the training-only use but forecasts a metric 4D state whose precision follows role, so the target's future pose is supervised explicitly.

Object-centric and geometric motion representations. Metric 3D inputs improve precision and sample efficiency [37, 62], and motion has been represented by point tracks [4, 52, 55], 3D flow [15, 18, 58, 61], object keypoints or SE(3) trajectories [17, 33, 41], and rigid-body decompositions [7, 8]. Each applies one representation to the whole scene, or models the target alone [14, 17]. DOCIR separates robot, target, and obstacles, but only as input encodings [13], and foveated robot perception allocates image resolution around a gaze point [11, 21, 23]. FoveaWAM allocates the precision of what is predicted: coarse sparse flow for the scene and a metric SE(3) trajectory for the target.

3. FoveaWAM

We present FoveaWAM, a native 4D world–action model that allocates precision by role on both sides of the present (Fig. 3). It encodes the past (Sec. 3.2) and predicts the future (Sec. 3.3) with the same split: a metric SE(3) trajectory for the target, coarse 3D geometry for its surroundings, and the current frame alone for everything static. An action stream reads this representation to generate time-indexed commands (Sec. 3.4).

3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query t, a policy π learned from demonstrations reads an RGB image I_t and a depth map D_t captured at time τ_t, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of H commands

$$(\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command j is meant for time $\tau_t + j\Delta_a$, with control period Δ_a.

Each task assigns the scene three roles. The *target* is the object whose pose the robot must match within a contact or containment tolerance. The *context* is the part of the surroundings that matters: obstacles, which the robot must clear, and drivers, which move the target. Everything else is *static*. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at τ_t.

3.2. Role-Aware Dynamics Encoding

What a policy must know about the past decouples by role. The static scene has no history worth encoding: the current frame already gives its layout. The context acts through the clearance it leaves, which is current geometry, and the motion it imparts, which shows in how the target has moved. Only the target's motion must be known precisely, and that motion is a rigid trajectory in SE(3). FoveaWAM therefore observes a single RGB-D frame for the scene and a short metric pose history for the target. Policies that take motion from stacked frames [10, 54] or optical-flow histories [14, 31] extract it densely over the whole image at every past step, and the target's velocity still remains implicit in appearance. FoveaWAM receives that velocity in physical units from a few low-dimensional tokens. This encoding keeps the model light enough to act within one control period while keeping the target's motion metric.

Static scene from the current frame. FoveaWAM anchors each patch token at a metric 3D point, so that the observation shares the base frame of the predictions and the actions. A frozen MAE ViT-B/16 [16] turns the RGB image into 196 patch tokens on a 14×14 grid, and the depth map back-projects the center of patch i to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. One frame thus gives the scene layout and its clearances as metric geometry. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

Target motion from the pose history. The target's recent motion enters as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units. The velocity and acceleration that decide where the target will be thus enter the model directly, and a driver such as a conveyor or a ramp is captured through the motion it has imparted to the target. Poses come from the simulator in simulation and from an online RGB-D tracker on the robot.

World stream. The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream. The entire motion history thus costs sixteen low-dimensional tokens, whereas each additional frame would cost 196.

3.3. Role-Aware 4D Future Prediction

Every command lands in a later scene than the one observed, so the representation must anticipate the scene at the execution times of the commands, and the precision this requires again follows role. Contact is decided by the target's pose, so FoveaWAM predicts the target's full SE(3) trajectory. The context matters through clearance and imparted motion, so FoveaWAM predicts the motion of the scene only coarsely,

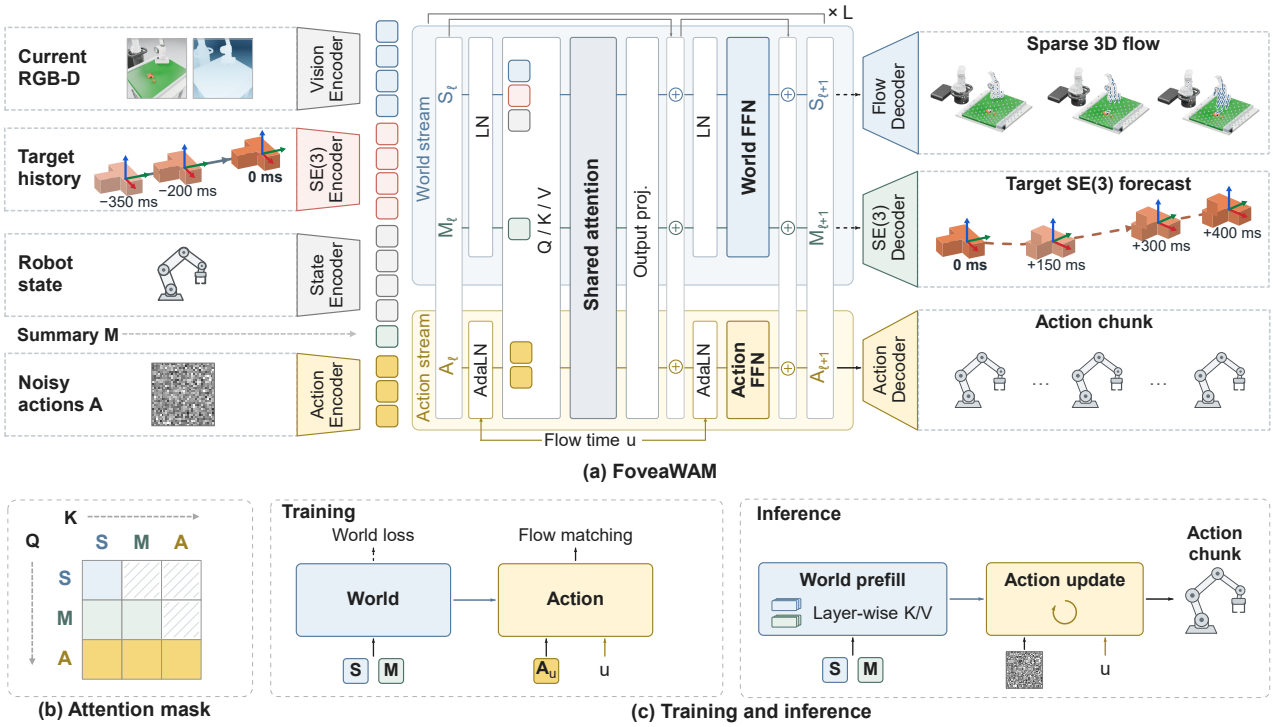


Figure 3. **Overview of FoveaWAM.** (a) Architecture. The current RGB-D frame and the target and robot histories form scene tokens S, alongside summary tokens M and noisy action tokens A. All tokens share masked attention, followed by separate world and action FFNs. The sparse 3D flow and the target SE(3) trajectory are predicted from S and M. (b) Attention mask. Rows are queries and columns are keys. (c) Training and inference. The world and action streams are trained jointly. At inference, the world stream runs once and the prediction heads are skipped.

as a sparse 3D flow on the patch grid. Both predictions are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50\,\text{ms}$ for $h = 1, \dots, 8$; together with the current state, these horizons cover the execution times of all commands in the chunk. Video and latent forecasts [49, 56] spend the same capacity on every part of the scene and leave the target's pose implicit. Our two predictions are linear readouts of tokens that the action stream already reads: they shape the representation during training and are skipped at inference, so prediction adds no latency.

Sparse 3D scene flow. At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for. For each of the 196 points $\mathbf{x}_i$ and each horizon h, a linear head on the final state of patch token i predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where F_{ih} is the displacement of the surface point at $\mathbf{x}_i$, o_{ih} its visibility, s a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

Target SE(3) trajectory. The flow samples the target only at the few patches it covers, too sparse to determine its rotation. A dedicated linear head on the mean of the final summary states therefore predicts its rigid motion directly: for each horizon h, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\phi}_h$, trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h\in\mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{3}$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t{+}\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\phi}_h^{\wedge}) R(\tau_t)$,

$d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. Contact demands the rotation term: a rotation error θ displaces a contact point at distance r from the target's origin by up to $2r\sin(\theta/2)$, nearly 9 mm for 5° at $r = 10$ cm.

3.4. World–Action Learning and Inference

Action stream. The action stream generates the chunk by conditional flow matching [27]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation increment, a rotation increment, and a gripper command. In every layer, one attention spans both streams with shared projections, followed by separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time. With S, M, and A denoting the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$, and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in X attend to tokens in $\mathcal{Y}$. The action stream thus reads, in every layer, both the patch states shaped by the scene flow and the summary states shaped by the target trajectory, while the world stream never reads the actions.

Training objective. For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\, \mathbf{a}_1 - \mathbf{a}_0\big), \tag{4}$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_{\mathrm{aux}} \mathcal{L}_{\mathrm{aux}}, \tag{5}$$

where $\mathcal{L}_F$ and $\mathcal{L}_P$ are the flow and pose losses of Eqs. (2) and (3), and $\mathcal{L}_{\mathrm{aux}}$ sums auxiliary appearance losses.

Inference. At deployment, FoveaWAM skips both prediction heads [25, 60, 63]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. One query takes 15.1 ms at p95 on one RTX 5090, within a single control period. Commands are indexed by time from capture, so the policy needs no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts from the command indexed to that boundary. Further implementation details are in the supplementary material.

4. The OnTimeBench Benchmark

4.1. Overview

Existing dynamic manipulation benchmarks stop the world while the policy thinks. The default evaluators of DOMINO [14] and PhysMani-Bench [61] advance the simulation only after the policy has returned its actions, so a slow policy is scored as if it were instant. Such evaluation measures what a policy predicts but not whether the prediction arrives in time, which is what decides success on a real robot.

To evaluate whether a policy can act on a moving scene in time, we introduce OnTimeBench, a benchmark of 14 dynamic manipulation tasks in which the world never pauses. Observation, scene motion, and action execution share one physical clock, so the scene keeps moving for as long as each policy computes. As shown in Fig. 4(a), a rolling ball travels 18 cm during a 200 ms inference, and the policy must act on where the ball will be. OnTimeBench covers five task families (Fig. 4(b)): *Grasp*, grasping a target under harmonic, conveyor, translating, rotating, or occluded motion; *Place*, placing a held object into a translating or rotating receptacle; *Pass*, passing a wand along a rotating wire; *Capture*, capturing a rolling or sliding object; and *Impact*, grasping a part after an impact. Fig. 4(c) shows all 14 tasks. Together they vary what decides success: the target's speed, from 0.03 to 1.14 m/s, its rotation, the clearance of obstacles, and the motion imparted by drivers such as conveyors, ramps, and strikers.

4.2. Construction

OnTimeBench is built in MuJoCo and robosuite [47, 69] with a Franka Panda arm controlled at 20 Hz. Each scene samples the target's initial state and motion parameters within task-specific ranges. A task-specific expert with privileged simulator state provides 800 demonstrations per task, 11,200 in total. All results are reported on 1,400 held-out test scenes, 100 per task, shared by all methods; configurations are selected on separate development scenes.

4.3. Evaluation Protocol

Native latency. During each query, physics advances by the method's own inference latency, drawn from its latency distribution measured at batch size 1 on one RTX 5090. All methods share one executor, which hands over each chunk at the first control-period boundary after inference completes and executes it from the command indexed to that boundary.

Settings and metrics. We evaluate three settings: *Native*, at each method's own latency; *+150 ms*, with 150 ms added to every method; and *Paused*, with physics paused during inference as in existing benchmarks. We report the task-balanced success rate (SR) and the p95 time-to-action, the

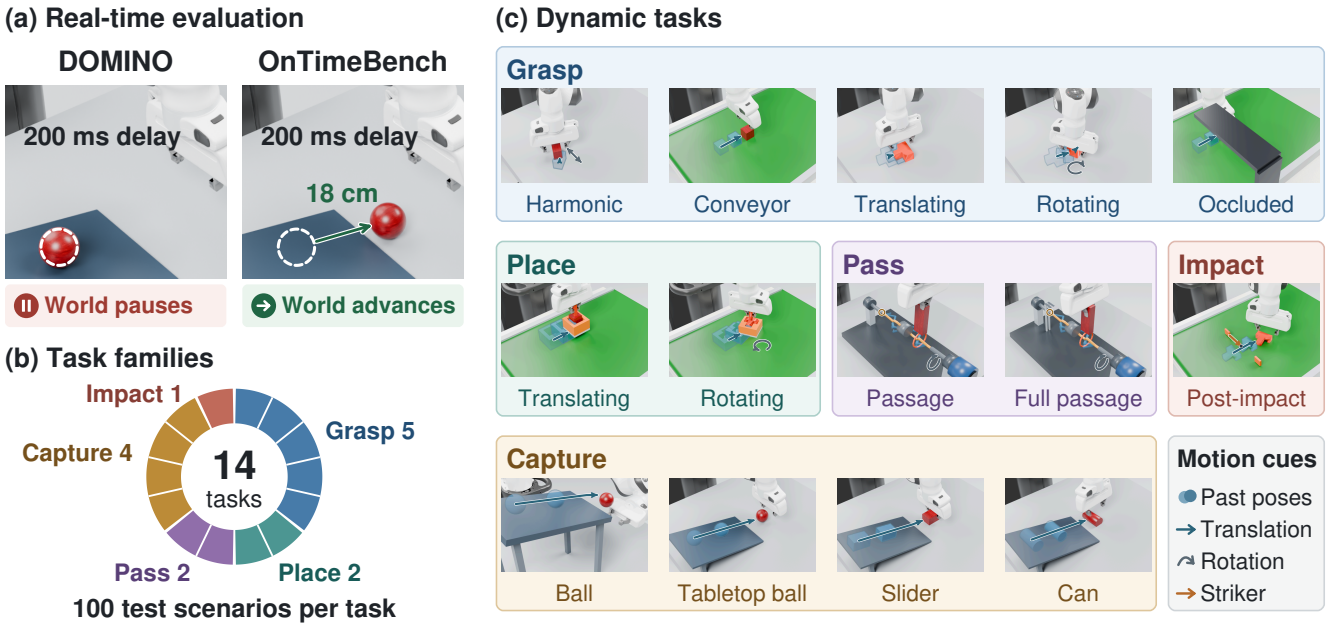


Figure 4. **Overview of OnTimeBench.** OnTimeBench evaluates each policy at its native latency, with 14 dynamic tasks across five families and 1,400 held-out test scenes. (a) Unlike DOMINO [14], which pauses the world during inference, OnTimeBench keeps it moving. (b) Distribution of task families. (c) Task demonstration.

time from an observation to its first command. The full protocol is in the supplementary material.

5. Experiments

5.1. Implementation Details

Baselines. We compare with the reactive policies ACT [66] and Diffusion Policy (DP) [10], the VLA $\pi_{0.5}$ [5], the dynamic VLAs DynamicVLA [54] and PUMA [14], the WAM OpenWAM-α [49], and the dynamic WAM DynamicWAM [31]. Each baseline keeps its official architecture and training recipe and is trained on the same demonstrations. To isolate the effect of target-motion input, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

Model and training. The transformer of FoveaWAM has 12 layers of width 1024 with 16 heads and 271.8M trainable parameters on top of the frozen MAE ViT-B/16 encoder. On OnTimeBench, it is trained from random initialization with AdamW for 10 epochs (25,870 updates). Inference runs 5 Euler steps in compiled BF16. FoveaWAM observes one agent-view RGB-D frame and the target's SE(3) pose history from the simulator.

5.2. Dynamic Manipulation Results

Results on OnTimeBench. As shown in Table 1, FoveaWAM achieves 82.1% SR at native latency, 20.7 points above the strongest baseline, PUMA (61.4%), and the highest SR in all three settings. It has the highest or tied-highest SR on 13 of the 14 tasks (Table S4), with the largest margins where a moving receptacle or the target's orientation decides success: 30 and 55 points on placement into trans-

Table 1. **Quantitative comparison on OnTimeBench.** SR (%) at each method's native latency, with 150 ms added, and with the world paused during inference. p95: time-to-action on one RTX 5090. Wrist marks baselines that use wrist RGB in at least one original setup. Baseline rollouts use only the main view. History: inputs that carry past motion. (off): caching and compilation disabled. Best and second best are highlighted.

Method	History				SR (%) ↑			p95 (ms) ↓
	Wrist RGB	Flow	SE(3)		Native	+150 ms	Paused	
ACT [66]	✓				41.6	24.6	40.6	**5.8**
DP [10]	✓	✓			31.9	16.1	32.0	68.9
$\pi_{0.5}$ [5]	✓				54.4	28.1	66.6	59.6
$\pi_{0.5}$+SE(3)	✓			✓	58.7	28.0	76.0	61.2
DynamicVLA [54]	✓	✓			51.1	23.5	54.9	97.3
PUMA [14]	✓		✓		61.4	31.1	53.1	60.6
OpenWAM-α [49]	✓				46.1	24.8	58.7	151.7
OpenWAM-α (off)	✓				15.8	10.4	58.4	466.3
DynamicWAM [31]	✓		✓		41.1	20.6	49.6	149.6
DynamicWAM+SE(3)	✓		✓	✓	41.7	24.9	64.0	153.3
FoveaWAM				✓	**82.1**	**34.1**	**90.3**	15.1

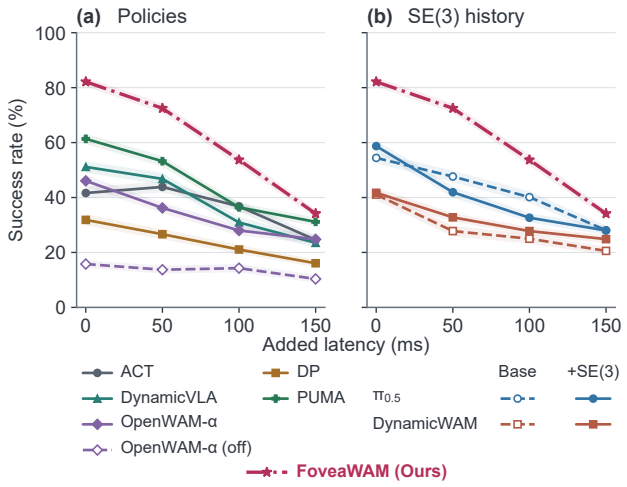


Figure 5. **Success under added latency.** SR on OnTimeBench as delay is added to each method's native latency. (a) Baselines. (off): caching and compilation disabled. (b) Base versus +SE(3) variants. Bands: 95% intervals.

lating and rotating receptacles, and 28 points on full wire passage. Precise target motion and low latency are both required for this result. Given the same SE(3) history, $\pi_{0.5}$ and DynamicWAM reach only 58.7% and 41.7%, and ACT, the only faster method, reaches 41.6%.

Effect of inference latency. Latency alone can decide success. The two OpenWAM-α deployments run the same policy and agree with the world paused (58.7% and 58.4%), yet at native latency the one with a p95 of 151.7 ms reaches 46.1% and the one with 466.3 ms reaches 15.8%. Latency also erodes the value of motion input: the SE(3) history raises $\pi_{0.5}$ and DynamicWAM by 9.4 and 14.4 points with the world paused but by only 4.3 and 0.6 at native latency. Paused evaluation therefore overestimates slow policies and misorders methods: $\pi_{0.5}$+SE(3), the strongest baseline with the world paused (76.0%), loses 17.3 points at native latency and falls behind PUMA. FoveaWAM leads in both settings and remains the best method at every added delay (Fig. 5).

Results on DOMINO. To evaluate FoveaWAM beyond OnTimeBench, we train FoveaWAM, $\pi_{0.5}$, DynamicWAM, and PUMA on the 24 single-arm Franka tasks of DOMINO [14] with its demonstrations and its native evaluator. As shown in Table 2, FoveaWAM achieves the highest overall SR, 46.8%, 19.9 points above the strongest baseline, DynamicWAM (26.9%), and the highest manipulation score. This evaluator pauses physics during inference, so the lead reflects the quality of the role-aware representation rather than its speed. The margin is largest on the 16 Track tasks, 55.5% against 31.9% for DynamicWAM.

Table 2. **Quantitative comparison on DOMINO** [14]. SR (%) on its 24 single-arm tasks, 8 Intercept and 16 Track. MS: manipulation score. Best and second best are highlighted.

Method	Intercept	Track	All ↑	MS ↑
$\pi_{0.5}$	23.8	8.1	13.3	27.8
DynamicWAM	16.8	31.9	26.9	41.4
PUMA	**36.3**	12.2	20.2	35.4
FoveaWAM	29.4	**55.5**	**46.8**	**62.0**

Table 3. **Ablation results on OnTimeBench.** SR (%) of FoveaWAM and its change for each variant (Δ, points; red: drop, green: gain). Rot./Ctx.: rotation-sensitive and context-active tasks.

	Success rate, all tasks		Groups (Native)	
	Native	+150 ms	Rot.	Ctx.
FoveaWAM	**82.1**	**34.1**	**80.4**	**82.7**
Precision allocation				
Uniform coarse flow	−4.5	−0.5	−8.6	−4.6
Uniform fine flow	+0.8	−0.9	−3.4	−2.6
w/o flow	−4.2	−1.4	−11.7	−3.9
Target motion history				
w/o history	−19.6	+0.2	−24.0	−2.3
Future supervision				
w/o future	−4.3	−3.4	−12.6	−2.1
w/o $\mathcal{L}_{\text{aux}}$	−2.6	−1.9	−11.6	+0.1

Table 4. **Real-robot results.** SR (%) over 20 trials per task. p95: end-to-end time-to-action (ms). Best is highlighted.

Method	Belt grasp	Box place	Turntable	Toy car grasp	Tennis ball	Ramp grasp	All ↑	p95 (ms) ↓
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
ACT	—	—	—	—	—	—	—	—
FoveaWAM	—	—	—	—	—	—	37.5	—

5.3. Ablation Study

We ablate each part of the role-aware design on OnTimeBench (Table 3) and report SR at native latency, overall and on the rotation-sensitive (Rot.) and context-active (Ctx.) task groups (Table S2). All variants except *w/o history* change only the training supervision, so they run at the same latency as FoveaWAM.

Effect of role-aware prediction. Role-aware prediction gives the target a precision that uniform prediction does not. Replacing the SE(3) trajectory with uniform flow at the same density reduces SR by 4.5 points overall and by 8.6 on rotation-sensitive tasks. Raising the uniform flow to a 56×56 grid, $16\times$ the points, costs [X.X]$\times$ the training time of FoveaWAM; it recovers the overall SR (+0.8) but still trails by 3.4 points on rotation-sensitive tasks and by 2.6

on context-active tasks. Removing the sparse flow instead reduces SR by 4.2 points overall, by 3.9 on context-active tasks, and by 11.7 on rotation-sensitive tasks. The two predictions are thus complementary: one target trajectory and 196 flow points match 3,136 uniform flow points overall and surpass them where rotation and context matter, at a fraction of the training cost.

Effect of target motion history. Removing the pose history while keeping the current target pose reduces SR by 19.6 points overall and by 24.0 on rotation-sensitive tasks, the largest drop among all variants. A single pose does not show how the target moves; the metric history supplies the motion that the commands must anticipate.

Effect of future supervision. Supervising only the current target state instead of its future reduces SR by 4.3 points overall and by 12.6 on rotation-sensitive tasks, although the prediction heads never run at deployment. Removing the auxiliary objectives $\mathcal{L}_{\text{aux}}$ reduces SR by 2.6 points overall and by 11.6 on rotation-sensitive tasks.

5.4. Real-Robot Evaluation

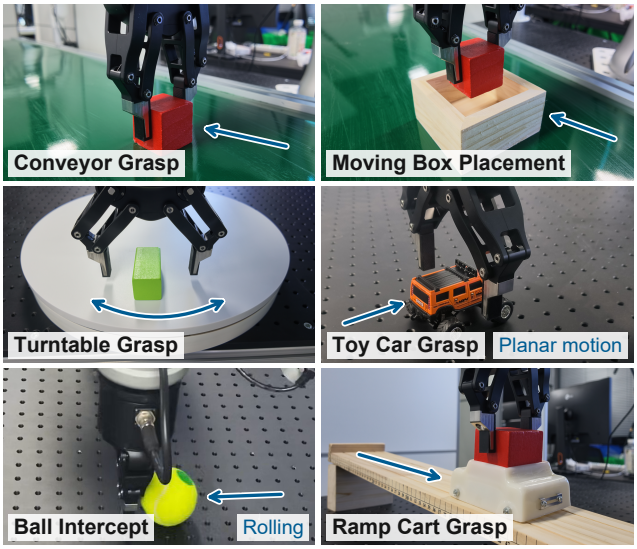


Figure 6. **Real-robot tasks.** Arrows show the motion of the target or the moving box before contact.

Setup. We evaluate on six dynamic tasks with a Flexiv Rizon 4s arm with a GN01 gripper (Fig. 6 and Table S8). All methods are trained on a shared set of 600 teleoperated demonstrations and evaluated for 20 trials per task. FoveaWAM starts from a DROID-pretrained base [22] and uses RGB-D and online target pose estimates from one fixed camera. The baselines, $\pi_{0.5}$, ACT, and DynamicWAM, use RGB from the fixed and wrist cameras.

Results. As shown in Table 4, FoveaWAM succeeds in 37.5% of all trials, against [XX.X]% for [the strongest baseline], with the largest margin on [tasks]. Its p95 latency, including tracking, is [XX] ms, against [XX] ms for [name]. [One trajectory-verified contact example: how the target's pose, rotation, or context explains a success of FoveaWAM or a failure of a baseline.]

6. Conclusion

We introduce FoveaWAM, a native 4D world–action model that addresses a tension at the core of dynamic manipulation: predicting a moving scene takes time, and that time lets the scene move further. By allocating precision by role, with a metric SE(3) trajectory for the target, sparse 3D flow for its surroundings, and the current frame for everything static, FoveaWAM is precise where contact is decided and acts in 15.1 ms. Together with OnTimeBench, which evaluates every policy at its native latency, it offers a model and a benchmark for building and evaluating policies that act on time.

LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

References

[1] Peter K. Allen, Aleksandar Timcenko, Billibon Yoshimi, and Paul Michelman. Automated tracking and grasping of a moving object with a robotic hand-eye system. *IEEE Transactions on Robotics and Automation*, 9(2):152–165, 1993. 2

[2] Yaakov Bar-Shalom, X. Rong Li, and Thiagalingam Kirubarajan. *Estimation with Applications to Tracking and Navigation: Theory, Algorithms and Software*. Wiley, 2001.

[3] Berthold Bäuml, Thomas Wimböck, and Gerd Hirzinger. Kinematically optimal catching a flying ball with a hand-arm-system. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2010. 2

[4] Homanga Bharadhwaj, Roozbeh Mottaghi, Abhinav Gupta, and Shubham Tulsiani. Track2Act: Predicting point tracks from internet videos enables generalizable robot manipulation. In *European Conference on Computer Vision*, 2024. 3

[5] Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. In *Proceedings of The 9th Conference on Robot Learning*, pages 17–40, 2025. 6, 14

[6] Kevin Black, Manuel Galliker, and Sergey Levine. Real-time execution of action chunking flow policies. In *Advances in Neural Information Processing Systems*, pages 33383–33407, 2025. 2, 13

[7] Arunkumar Byravan and Dieter Fox. SE3-Nets: Learning rigid body motion using deep neural networks. In *IEEE International Conference on Robotics and Automation*, pages 173–180, 2017. 3

[8] Arunkumar Byravan, Felix Leeb, Franziska Meier, and Dieter Fox. SE3-Pose-Nets: Structured deep dynamics models for visuomotor control. In *IEEE International Conference on Robotics and Automation*, 2018. 3

[9] Chongkei Chang and Zhidong Deng. DynaWM: A base-VLA-guided world foundation model for moving-object manipulation. *arXiv preprint arXiv:2607.02604*, 2026. 2

[10] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 3, 6, 14

[11] Ian Chuang, Jinyu Zou, Andrew Lee, Dechen Gao, and Iman Soltani. Look, focus, act: Efficient and robust robot learning via human gaze and foveated vision transformers. In *IEEE International Conference on Robotics and Automation*, pages 18506–18513, 2026. 3

[12] Christine A. Curcio, Kenneth R. Sloan, Robert E. Kalina, and Anita E. Hendrickson. Human photoreceptor topography. *Journal of Comparative Neurology*, 292(4):497–523, 1990. 2

[13] David Emukpere, Romain Deffayet, Bingbing Wu, Romain Brégier, Michael Niemaz, Jean-Luc Meunier, Denys Proux, Jean-Michel Renders, and Seungsu Kim. Disentangled object-centric image representation for robotic manipulation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 15873–15879, 2025. 3

[14] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. In *European Conference on Computer Vision*, pages 314–333, 2026. 1, 2, 3, 5, 6, 7, 14, 15

[15] Jun Guo, Xiaojian Ma, Yikai Wang, Min Yang, Huaping Liu, and Qing Li. FlowDreamer: A RGB-D world model with flow-based motion representations for robot manipulation. *IEEE Robotics and Automation Letters*, 11(3):2466–2473, 2026. 3

[16] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022. 3, 12

[17] Cheng-Chun Hsu, Bowen Wen, Jie Xu, Yashraj Narang, Xiaolong Wang, Yuke Zhu, Joydeep Biswas, and Stan Birchfield. SPOT: SE(3) pose trajectory diffusion for object-centric manipulation. In *IEEE International Conference on Robotics and Automation*, 2025. 1, 2, 3

[18] Wenlong Huang, Yu-Wei Chao, Arsalan Mousavian, Ming-Yu Liu, Dieter Fox, Kaichun Mo, and Li Fei-Fei. PointWorld: Scaling 3D world models for in-the-wild robotic manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20765–20779, 2026. 3, 12

[19] Roland S. Johansson, Göran Westling, Anders Bäckström, and J. Randall Flanagan. Eye–hand coordination in object manipulation. *Journal of Neuroscience*, 21(17):6917–6932, 2001. 2

[20] Nikita Karaev, Iurii Makarov, Jianyuan Wang, Natalia Neverova, Andrea Vedaldi, and Christian Rupprecht. CoTracker3: Simpler and better point tracking by pseudo-labelling real videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. 16

[21] Justin Kerr, Kush Hari, Ethan Weber, Chung Min Kim, Brent Yi, Tyler Bonnen, Ken Goldberg, and Angjoo Kanazawa. Eye, robot: Learning to look to act with a BC-RL perception-action loop. In *Proceedings of The 9th Conference on Robot Learning*, pages 3647–3664, 2025. 3

[22] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. In *Proceedings of Robotics: Science and Systems*, 2024. 8, 12

[23] Heecheol Kim, Yoshiyuki Ohmura, and Yasuo Kuniyoshi. Gaze-based dual resolution deep imitation learning for high-precision dexterous robot manipulation. *IEEE Robotics and Automation Letters*, 6(2):1630–1637, 2021. 3

[24] Michael F. Land and Peter McLeod. From eye movements to

actions: How batsmen hit the ball. *Nature Neuroscience*, 3 (12):1340–1345, 2000. 2

[25] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model. In *Proceedings of Robotics: Science and Systems*, 2025. 2, 5

[26] Haoran Liao, Pengyue Wang, Shuoyu Chen, Kehan Cheng, Xuhang Chen, Yuhao Lin, Mu Lin, Zhizhao Liang, Xiaoyi Fan, Chengyi Xing, Dan Niu, Yi-Lin Wei, and Wei-Shi Zheng. DynamicManip: Enabling dynamic manipulation from a single static demonstration. *arXiv preprint arXiv:2608.01452*, 2026. 2

[27] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. 5

[28] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision*, 2024. 16

[29] Yuejiang Liu, Jubayer Hamid, Annie Xie, Yoonho Lee, Max Du, and Chelsea Finn. Bidirectional decoding: Improving action chunking via guided test-time sampling. In *International Conference on Learning Representations*, pages 4594–4627, 2025. 2

[30] Yushan Liu, Peibo Sun, Shoujie Li, Yifan Xie, Lingfeng Zhang, Xintao Chao, Shiyuan Dong, Fang Chen, Xiao-Ping Zhang, and Wenbo Ding. OA-WAM: Object-addressable world action model for robust robot manipulation. *arXiv preprint arXiv:2605.06481*, 2026. 2

[31] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 1, 2, 3, 6, 14

[32] Siyuan Ma, Yutian Zhang, Boshi Zhang, Qinglian Wu, Jiaqi Zhai, Dong Wei, and Xiaojin Huang. ForeTime-VLA: Causal future-token distillation from a world action model for conveyor-belt manipulation. *arXiv preprint arXiv:2608.20735*, 2026. 1, 2

[33] Lucas Manuelli, Wei Gao, Peter Florence, and Russ Tedrake. kPAM: Keypoint affordances for category-level robotic manipulation. In *Robotics Research: The 19th International Symposium ISRR*, pages 132–157. Springer, 2022. 3

[34] Naresh Marturi, Marek Kopicki, Alireza Rastegarpanah, et al. Dynamic grasp and trajectory planning for moving objects. *Autonomous Robots*, 43(5):1241–1256, 2019. 2

[35] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *International Conference on Learning Representations*, pages 28085–28128, 2025. 16

[36] Ruth Rosenholtz. Capabilities and limitations of peripheral vision. *Annual Review of Vision Science*, 2:437–457, 2016. 2

[37] Mohit Shridhar, Lucas Manuelli, and Dieter Fox. Perceiver-actor: A multi-task transformer for robotic manipulation. In *Conference on Robot Learning*, pages 785–799, 2023. 3

[38] Mustafa Shukor, Dana Aubakirova, Francesco Capuano, Pepijn Kooijmans, Steven Palma, et al. SmolVLA: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025. 2

[39] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 12

[40] Otto J. M. Smith. Closer control of loops with dead time. *Chemical Engineering Progress*, 53(5):217–219, 1957. 2

[41] Rustin Soraki, Homanga Bharadhwaj, Ali Farhadi, and Roozbeh Mottaghi. ObjectForesight: Predicting future 3D object trajectories from human videos. In *European Conference on Computer Vision*, pages 437–453, 2026. 3

[42] Hans Strasburger, Ingo Rentschler, and Martin Jüttner. Peripheral vision and pattern recognition: A review. *Journal of Vision*, 11(5):13, 2011. 2

[43] Yue Su, Sijin Chen, Haixin Shi, Mingyu Liu, Zhengshen Zhang, Ningyuan Huang, Weiheng Zhong, Zhengbang Zhu, Yuxiao Liu, and Xihui Liu. World guidance: World modeling in condition space for action generation. In *Proceedings of the 43rd International Conference on Machine Learning*, pages 115917–115933, 2026. 2

[44] Shahram Najam Syed, Arthur Jakobsson, Haoran Hao, and Jeffrey Ichnowski. Intercepting the future: Latent-space predictive world model for dynamic VLA manipulation. In *Conference on Robot Learning*, 2026. Accepted for publication. 2

[45] Jiaming Tang, Yufei Sun, Yilong Zhao, Shang Yang, Yujun Lin, Zhuoyang Zhang, James Hou, Yao Lu, Zhijian Liu, and Song Han. VLASH: Real-time VLAs via future-state-aware asynchronous inference. *arXiv preprint arXiv:2512.01031*, 2025. 2

[46] Yang Tian, Sizhe Yang, Jia Zeng, Ping Wang, Dahua Lin, Hao Dong, and Jiangmiao Pang. Predictive inverse dynamics models are scalable learners for robotic manipulation. In *International Conference on Learning Representations*, pages 92033–92052, 2025. 2

[47] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033, 2012. 5, 13

[48] Ruicheng Wang, Sicheng Xu, Yue Dong, Yu Deng, Jianfeng Xiang, Zelong Lv, Guangzhong Sun, Xin Tong, and Jiaolong Yang. MoGe-2: Accurate monocular geometry with metric scale and sharp details. In *Advances in Neural Information Processing Systems*, pages 35928–35959, 2025. 13

[49] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An open, modular exploration towards systematic world-action model pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 1, 2, 4, 6, 14

[50] Bowen Wen, Wei Yang, Jan Kautz, and Stan Birchfield. FoundationPose: Unified 6D pose estimation and tracking of novel objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17868–17879, 2024. 16

[51] Bowen Wen, Matthew Trepte, Joseph Aribido, Jan Kautz, Orazio Gallo, and Stan Birchfield. FoundationStereo: Zero-shot stereo matching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5249–5260, 2025. 16

[52] Chuan Wen, Xingyu Lin, John Ian Reyes So, Kai Chen, Qi Dou, Yang Gao, and Pieter Abbeel. Any-point trajectory modeling for policy learning. In *Proceedings of Robotics: Science and Systems*, 2024. 3

[53] Hongtao Wu, Ya Jing, Chilam Cheang, Guangzeng Chen, Jiafeng Xu, Xinghang Li, Minghuan Liu, Hang Li, and Tao Kong. Unleashing large-scale video generative pre-training for visual robot manipulation. In *International Conference on Learning Representations*, pages 10641–10662, 2024. 2

[54] Haozhe Xie, Beichen Wen, Jiarui Zheng, Zhaoxi Chen, Fangzhou Hong, Haiwen Diao, and Ziwei Liu. DynamicVLA: A vision-language-action model for dynamic object manipulation. In *Advances in Neural Information Processing Systems*, 2026. Accepted for publication. 2, 3, 6, 14

[55] Mengda Xu, Zhenjia Xu, Yinghao Xu, Cheng Chi, Gordon Wetzstein, Manuela Veloso, and Shuran Song. Flow as the cross-domain manipulation interface. In *Conference on Robot Learning*, pages 2475–2499, 2025. 3

[56] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, et al. World action models are zero-shot policies. *arXiv preprint arXiv:2602.15922*, 2026. 1, 2, 4

[57] Hanyang Yu, Haitao Lin, Jingbo Zhang, Wenyao Zhang, Chenghao Gu, Heng Li, and Ping Tan. MaskWAM: Unifying mask prompting and prediction for world-action models. *arXiv preprint arXiv:2606.13515*, 2026. 2

[58] Chengbo Yuan, Chuan Wen, Tong Zhang, and Yang Gao. General flow as foundation affordance for scalable robot learning. In *Conference on Robot Learning*, pages 1541–1566, 2025. 3

[59] Shanglin Yuan, Weiheng Zhao, Xin Shi, Haoyi Jiang, Xianda Guo, Liu Liu, Wenyu Liu, Wei Sui, and Xinggang Wang. DreamWAM: Beyond RGB future prediction for world action models. *arXiv preprint arXiv:2608.04996*, 2026. 2

[60] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination? In *Advances in Neural Information Processing Systems*, 2026. Accepted for publication. 2, 5

[61] Peng Yun, Shouwang Huang, Hao Li, Jinxi Li, Jianan Wang, and Bo Yang. PhysMani: Physics-principled 3D world model for dynamic object manipulation. In *European Conference on Computer Vision*, pages 20–39, 2026. 1, 2, 3, 5

[62] Yanjie Ze, Gu Zhang, Kangning Zhang, Chenyuan Hu, Muhan Wang, and Huazhe Xu. 3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations. In *Proceedings of Robotics: Science and Systems*, 2024. 3

[63] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, XinQiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 2, 5

[64] Yifan Zhang, Ruiping Wang, and Xilin Chen. Dynamic behavior cloning with temporal feature prediction: Enhancing robotic arm manipulation in moving object tasks. *IEEE Robotics and Automation Letters*, 10(6):5209–5216, 2025. 1, 2

[65] Ruiteng Zhao, Zhengshen Zhang, Yue Su, Wenshuo Wang, Jiahui Li, Zhiyuan Yang, Francis E. H. Tay, Marcelo H. Ang, and Haiyue Zhu. SG-WAM: Self-guided world modeling in geometry-aware policy space. *arXiv preprint arXiv:2608.01397*, 2026. 2

[66] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems*, 2023. 2, 6, 14

[67] Ruijie Zheng, Jing Wang, Scott Reed, et al. FLARE: Robot learning with implicit world modeling. In *Proceedings of The 9th Conference on Robot Learning*, pages 3952–3971. PMLR, 2025. 2

[68] Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified world models: Coupling video and action diffusion for pretraining on large robotic datasets. In *Proceedings of Robotics: Science and Systems*, 2025. 2

[69] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. *arXiv preprint arXiv:2009.12293*, 2020. 5, 13

A. Method and Training Details

A.1. Architecture and Training

On OnTimeBench, each training window holds eight consecutive query positions, each with its own causal history (Table S1). We report the final checkpoint of the seed-0 run.

Table S1. Hyperparameters of FoveaWAM on OnTimeBench and training clusters used across the project.

Parameter	Value
Model	
Input image	224×224 RGB-D, one view
Visual encoder	MAE ViT-B/16, frozen
History length	8 steps (350 ms)
Forecast horizons	8 (50 ms–400 ms)
Summary tokens	16
Layers	12
Width	1024
Heads	16
Feed-forward width	4096
Trainable parameters	271.8M
Action	
Action dimension	7
Chunk length	8
Euler steps	5
Noise samples per query	8
Training	
Optimizer	Fused AdamW
Betas	$(0.9, 0.999)$
Weight decay	10^{-4}
Learning rate	2×10^{-4}
LR schedule	cosine, 5,174 warm-up updates
Batch size	512 windows
Epochs	10 (cap: 20)
Reported checkpoint	25,870 updates
Gradient clipping	0.1
$\lambda_F, \lambda_P, \lambda_R, \lambda_{\text{aux}}$	0.1
Training precision	BF16 compute, FP32 parameters
Training clusters	$64 \times$ RTX PRO 6000, $32 \times$ H100

Scene tokens. Each patch token of the frozen MAE ViT-B/16 [16] adds a learned view embedding and linear embeddings of its mean valid metric depth and of the base-frame position $\mathbf{x}_i$ of its back-projected center pixel; a patch without valid depth at its center receives no position embedding. Only the patch queries and keys receive 3D rotary embeddings from $\mathbf{x}_i$, and the same points are the source points of the sparse 3D flow.

State tokens. The 8 target-state tokens sample the offsets $-350, -300, \ldots, 0$ ms. Each is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$: slots 0–6 describe the seven intervals between consecutive offsets, with velocities and accelerations from finite differences over the elapsed times, and slot 7 holds the current pose in the base frame. Each dimension is standardized with training statistics, invalid dimensions are zeroed, and three validity flags (pose, interval, acceleration) are appended; a slot without any valid feature is masked out of attention. The robot tokens hold the end-effector position, rotation vector, and gripper opening at the same offsets, and 16 learned embeddings form the summary tokens.

Heads and losses. Each head is a linear layer on the final world-stream states, except the target reconstruction, a two-layer MLP. The pose head reads the mean summary state ($\mathcal{L}_P$, Eq. 3). The flow head gives each patch point's displacement and visibility logit at every horizon ($\mathcal{L}_F$, Eq. 2, with scale $s = 0.715$ m and visibility weight 1); valid but invisible points still supervise their displacement. Flow labels follow the rendered surface point at each patch center through the ground-truth rigid motion of its MuJoCo geometry; a future point is visible if it projects into the image onto the same geometry within a depth tolerance of $\max(1\,\text{mm}, 0.001z)$. $\mathcal{L}_{\text{aux}}$ sums four losses: squared errors on future metric depth (valid pixels), on the RGB of surfaces that move by more than 1 mm, and on layer-normalized DINOv3 features [39], and an absolute error on the current target's RGB region reconstructed from the mean summary state.

Action stream. Each action token embeds a seven-dimensional command: a base-frame translation increment, a left-relative rotation vector, and a gripper command, relative to the measured end-effector pose at application. Command j of query t is labeled with the demonstration command issued at $\tau_t + j\Delta_a$, since d_{task} applies only at execution (Eq. (S1)), and commands are standardized over the training split. At deployment, the gripper changes state only when its command leaves a deadband of 0.5 in Ball capture and 0.1 in every other task, including Ball capture, table.

A.2. Pretraining on DROID and Fine-Tuning

The real-robot base trains on DROID [22] (15 Hz) with the first exterior camera and its depth, using the 3D point tracks and calibrations released by PointWorld [18]. Its 42,087 usable episodes yield 598,889 anchor windows, 60,149 of them for validation; each anchor expands into 8 query positions as on OnTimeBench, and only the anchor carries flow and DINOv3 labels. Each patch follows one PointWorld track that starts inside it, preferring the one with the largest consistent motion so that small moving objects are covered; 47% of the point–horizon slots carry a label.

DROID provides no object poses, so the base omits the target-state tokens and pose head. The losses for actions, sparse 3D flow, and future DINOv3 features have weights 1, 0.1, and 0.1, respectively. The flow loss has no visibility term. We cap pretraining at 20 epochs (21,060 optimizer updates)

with AdamW and a global batch size of 512. The backbone uses FP32 and the frozen visual encoder uses BF16.

Each real-robot policy starts from a DROID-pretrained base and is fine-tuned on its task's demonstrations, with target-state tokens and new prediction heads initialized from scratch. The fine-tuning budget is 20,000 optimizer updates per task with a global batch of 32. AdamW uses 500 warm-up updates followed by cosine decay, with peak learning rates of 3×10^{-5} and 2×10^{-5} for the 12- and 24-layer bases, respectively.

A.3. Figure Details

In Fig. 1(a), the real-robot scene is reconstructed from RGB with MoGe-2 [48], and the overlaid SE(3) futures and sparse 3D flow are illustrative. The overlays of Fig. 2 come from one recorded OnTimeBench scene: flow samples with a displacement of at most 2 mm are drawn as dots, and the target axes are drawn at 0 and 400 ms. In Fig. 3, the thumbnails are Blender renderings of a recorded rollout with the flow enlarged for display; token counts, layer stacking, and action icons are schematic. Fig. 4(c) renders expert rollouts with past target poses and motion cues.

B. OnTimeBench and Evaluation Protocol

B.1. Benchmark Design

The tasks run in MuJoCo 3.11.0 and robosuite 1.5.2 [47, 69] with a Franka Panda arm and a parallel gripper under robosuite's operational-space pose controller, with a control period of 50 ms and a physics step of 2 ms. A fixed third-person camera with a 40° vertical field of view renders 224×224 RGB-D images. An action is an end-effector translation and rotation-vector increment in the base frame with a gripper command, clipped to 0.10 m per translation axis and 0.5 rad per rotation component. Table S2 lists the fourteen tasks.

B.2. Scenarios, Demonstrations, and Splits

Each scene samples the target's initial state and motion parameters uniformly within task-specific ranges. Table S2 summarizes target motion. Other sampled parameters include harmonic amplitudes of up to 35 mm at 0.25–0.7 Hz, yaw rates of up to 1.2 rad/s, and ramp slopes of 5–15°. In an occluded scene, the target is hidden for 0.5–3.0 s and then reappears. Only physically feasible scenes are kept, and evaluation replays the stored scenes.

The expert is a task-specific closed-loop state machine with privileged simulator state that anticipates the target's motion, for example by predicting the interception point in the capture tasks. A demonstration is kept if the expert succeeds, a replay of its actions reproduces the episode, and its path and jerk pass quality checks. The training pool holds 11,200 demonstrations, 800 per task, recorded at 20 Hz with RGB-D, robot states, commands, and simulator ground truth, and 350 further episodes form the validation split; policies receive only the inputs of Sec. 3.

We select configurations on 1,400 development scenes (100 per task) and report all OnTimeBench results on 1,400 held-out scenes shared by all methods, generated with the same ranges from seeds used in no training, validation, or development episode.

B.3. Real-Time Evaluation and Metrics

Executor. All methods share one executor on the 50 ms control clock. Command j of query t is meant for $\tau_t + j\Delta_a$, and its execution time (Sec. 3.1) is

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + j\Delta_a + d_{\mathrm{task}}, \tag{S1}$$

where $d_{\mathrm{task}} = 50$ ms for the post-impact grasp, which applies each command 50 ms after it is issued, and $d_{\mathrm{task}} = 0$ otherwise. As in RTC [6], latency counts in whole control periods: a query with total latency L_t hands over its chunk at the first boundary at or after it ends, $k_t = \lceil L_t/\Delta_a \rceil$ periods after capture, and execution continues from command k_t, the one meant for that boundary. The policy then observes again and starts the next query; meanwhile the previous chunk continues, so a policy replans as often as its latency allows. A chunk holds the method's native number H_m of commands (Table S3) and is dropped if $k_t \geq H_m$. Before the first chunk arrives and whenever a chunk runs out, the controller holds the current arm target and gripper command; no command is repeated. The executor runs asynchronously, as in the streaming of DynamicVLA; no method, including FoveaWAM, inpaints across chunks as in RTC.

Protocols. In real time, physics advances during each query by the method's latency, drawn per query from its distribution measured on one RTX 5090, while the rollouts themselves run on RTX PRO 6000. With paused physics, inference takes no physics time and a policy replans every 50 ms; this removes the wait for inference but also raises the feedback rate, so it is a reference rather than an upper bound. The latency sweep adds 0 to 200 ms in steps of 50 ms after inference in real time.

Latency measurement. We time each method alone on one RTX 5090 at batch size 1 in each baseline's release precision and in compiled BF16 for FoveaWAM, over three rounds of 30 warm-up and 300 measured calls on recorded observations with the required causal history. A timed query starts from the newest observation and ends when its first command is available. It includes all computation that needs the newest frame, such as preprocessing, image encoding, the online flow and motion descriptors of PUMA and DynamicWAM, sampling, and action decoding, and on the robot also tracking. PUMA is timed with its official three-camera

Table S2. **The fourteen OnTimeBench tasks.** The target is the body whose pose the policy receives (Sec. 3.1). For placement it is the receptacle. For passage it is the wire entry, and the rest of the wire is context. Motion gives linear speed in m/s or the wire's angular speed in degrees per second. For each capture task, we estimate speed from the two samples 50 ms apart that bracket the first recorded ramp exit in each of 100 held-out fixed-robot reference trajectories. A grasp requires bilateral fingerpad contact, a lift of at least 0.05 m, and a continuous 0.5 s hold. An episode fails at unsafe contact, leaving the workspace, a drop, or the time limit. In the group column (Table 3), R marks tasks where the target rotates during the lead and its orientation matters at contact. C marks tasks with an obstacle clearance or a driver such as a conveyor, ramp, or striker.

Task	Code	Target	Motion range	Success	Limit (s)	Group
Harmonic grasp	`grasp`	block	0.03–0.20	grasp	8	R
Conveyor grasp	`t1_belt`	part	0.04–0.10	grasp	8	C
Translating grasp	`t1_xyz`	part	0.04–0.15	grasp	8	–
Rotating grasp	`t1_so3`	part	0.04–0.15	grasp	8	R
Occluded grasp	`t1_occ`	part	0.04–0.10	grasp without canopy contact	8	C
Moving receptacle	`t2_trans`	receptacle	0.03–0.06	held object at rest inside	10	–
Rotating receptacle	`t2_se3`	T-shaped receptacle	0.03–0.06	held T-part at rest inside	10	R
Wire passage	`t3_phase`	wire entry	2.5–7.5°/s	align within 30 mm, pass without contact	21	R, C
Full wire passage	`t3_full`	wire entry	2.5–7.5°/s	grasp wand, pass, exit, release	32	R, C
Ball capture	`t4_sphere`	ball	0.67–1.14	grasp before the tray	6	C
Ball capture on table	`t4_sphere_tabletop`	ball	0.59–1.03	grasp while rolling	6	C
Slider capture	`t4_slider`	slider	0.55–1.12	grasp while sliding	6	C
Can capture	`t4_can`	can	0.59–0.97	grasp before tipping	6	R, C
Post-impact grasp	`t5_replan`	part	0.12–0.14	grasp along its axis	6	R, C

input and the other baselines with two cameras. FoveaWAM runs five Euler steps with TF32 disabled; compilation and warm-up on recorded observations finish before physics starts, after which the history and the inference seed are reset. The forecast heads of FoveaWAM are skipped at deployment and excluded from time-to-action. A query misses its deadline when this time-to-action exceeds 50 ms, and the action age of a command is its execution time minus the capture time. Appendix C.2 reports the measured values.

Metrics. SR weights the 14 tasks equally. The position error at first contact e_p is the distance of the end effector from the expert's contact frame on the target; in the placement tasks, which start with the object held, it uses the contact of the held object with the receptacle and the expert's placement pose. e_p averages over the 13 tasks with a contact, excluding `t3_phase`, and over the episodes that make contact; failures without contact remain in the SR denominator. Paired 95% intervals bootstrap held-out scenes within each task, jointly across methods.

B.4. Baselines

Common conditions. Each baseline keeps its official architecture, chunk length, initialization, and training recipe (Table S3), trains on the same 11,200 demonstrations mapped to its action space through the same Cartesian controller, and is configured on development scenes. Rollouts use the agent view each baseline was trained with, any wrist slot masked, and the histories of Table 1; masking the wrist view changes p95 latency by at most 3.8 ms in the seven configurations timed both ways. DP uses its authors' real-robot DDIM sampler with 16 steps, and OpenWAM-α runs with video decoding disabled and is timed with and without its official cache and compilation.

Table S3. **Baseline training.** The training budget gives optimizer updates and global batch size. H is the number of commands returned per query. The parameter counts include trainable parameters only. DOM ckpt. denotes the released DynamicVLA checkpoint trained on DOM.

Method	Initialization	Updates×batch	H	Params
ACT [66]	scratch	50,000 × 32	100	83.9M
DP [10]	scratch	50,000 × 32	15	288.7M
$\pi_{0.5}$ [5]	`pi05_base`	30,000 × 256	10	3,353.4M
$\pi_{0.5}$+SE(3)	`pi05_base`	30,000 × 256	10	3,357.7M
DynamicVLA [54]	DOM ckpt.	50,000 × 44	20	99.1M
PUMA [14]	official	40,000 × 64	16	4,516.5M
OpenWAM-α [49]	official	34,490 × 192	32	6,021.2M
DynamicWAM [31]	official	40,000 × 128	16	988.8M
DynamicWAM+SE(3)	official	40,000 × 128	16	989.4M
FoveaWAM	scratch	25,870 × 512	8	271.8M

SE(3)-augmented baselines. $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Appendix A.1 from the same pose source, with the same validity flags. A two-layer MLP with a zero-initialized output layer maps them to the model's width; for $\pi_{0.5}$ they follow the image tokens in the prefix, where the action expert attends to them, and for DynamicWAM they join its kinematic descriptors. Training follows the base model's recipe.

C. Additional Simulation Results

C.1. Per-Task Results

Table S4 uses the Native real-time protocol with +0 ms extra delay. FoveaWAM reaches 93% on moving-receptacle placement and 66% on rotating-receptacle placement, compared with 63% and 11% for $\pi_{0.5}$+SE(3). On full wire passage, FoveaWAM reaches 40% versus 12% for DynamicVLA. It ties the strongest baseline on phase-only wire passage, tabletop ball capture and can capture. The largest recorded failure classes for FoveaWAM are rim impulse violations (160 of 1,400 scenes) and wire contacts (55 scenes).

C.2. Latency

Table S5 lists all eleven configurations at five added latencies; Fig. 5 plots the points from 0 to 150 ms. At +200 ms the 8-command chunk of FoveaWAM and the 10-command chunk of $\pi_{0.5}$ no longer cover the interval until the next chunk arrives: three and four commands remain for five and six control periods, and the controller holds its last target in between. That column therefore measures chunk length rather than the response to delay.

Table S5. **Success versus added latency** on 1,400 held-out OnTimeBench scenes per point (SR in %, real-time protocol). The 0 and 150 ms columns equal the Native and +150 ms columns of Table 1. Blue/bold and light blue mark the best and second-best values per column.

	Added latency (ms)				
Method	0	50	100	150	200
ACT	41.6	43.9	36.8	24.6	16.6
Diffusion Policy	31.9	26.6	21.1	16.1	14.4
$\pi_{0.5}$	54.4	47.6	40.1	28.1	13.6
$\pi_{0.5}$ + SE(3)	58.7	41.9	32.6	28.0	14.9
DynamicVLA	51.1	46.8	30.9	23.5	18.0
PUMA	61.4	53.2	36.4	31.1	**27.2**
OpenWAM-α	46.1	36.2	28.0	24.8	22.1
OpenWAM-α (off)	15.8	13.7	14.3	10.4	9.5
DynamicWAM	41.1	27.8	25.0	20.6	21.0
DynamicWAM + SE(3)	41.7	32.8	27.8	24.9	22.9
FoveaWAM	**82.1**	**72.5**	**53.7**	**34.1**	12.8

Timing. Table S6 reports replay timings by stage for the development checkpoint at 20,696 updates.

Table S6. **Latency breakdown** of one query of compiled BF16 FoveaWAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot. The policy total uses 900 measured queries and excludes pose tracking; magenta dashes mark pending measurements.

Stage	p50	p95
Pose tracking (per frame, robot)	—	—
Observation encoding + H2D	4.83	5.28
World stream (once per query)	3.93	4.22
Action stream (5 Euler steps)	4.27	4.41
Policy time-to-action (total)	14.66	15.65
Training-only readouts (for reference)	0.37	0.41
Action age at execution (robot)	—	—
Queries over 50 ms (robot, %)	—	

C.3. Results on DOMINO

We use the 24 single-arm Franka tasks of the clean level-1 setting of DOMINO [14] with its demonstrations, scenes, and native evaluator. The manipulation score (MS) scales route completion by penalties for leaving the workspace or view and for collisions. FoveaWAM reads the ground-truth pose of the instructed target from the simulator and masks slots without a valid pose (Appendix A.1). The RGB-based baselines do not receive this pose.

Data and adaptation. The data split has 1,200 training and 240 validation demonstrations. Policies execute absolute joint-position targets with DOMINO's native controller. Their state and action projections use its eight-dimensional joint space, with normalization fitted to the training data. The official sampler of DynamicWAM skips demonstrations too short for its windows, which leaves it 1,054 training and 210 validation demonstrations and none for two tasks; all 24 tasks remain in the test set. DynamicWAM trains for 40,000 updates at global batch 128. $\pi_{0.5}$ starts from its official base, trains for 30,000 updates at global batch 256, and is evaluated with its final EMA. PUMA keeps its published weights and frozen teacher and trains at global batch 64 with early stopping on validation action MSE, which selected update 6,000. For development, FoveaWAM trains from scratch with a global batch of 512 and stops after 1,575 updates. The lowest validation loss selects the checkpoint at update 1,365. Development rollouts select three executed actions per query.

C.4. Ablation Controls

The six controls of Sec. 5.3 (Table 3) share the data, training recipe, and executor. Each is evaluated at the final checkpoint of its run. Removing a loss also removes its head. The control without target history removes slots 0–6 and keeps slot 7, the current pose (Appendix A.1). The uniform fine

Table S4. **Per-task held-out success** (%) on OnTimeBench for every method of Table 1, Native real-time protocol: each method's measured inference latency plus **+0 ms extra delay**, with physics continuing during inference; 100 paired scenes per task. FoveaWAM uses compiled BF16 C (p95 15.1 ms). DP: Diffusion Policy. DVLA: DynamicVLA. OWAM: OpenWAM-α with official cache/compile; OW-off: both disabled. DWAM: DynamicWAM. +: with our SE(3) history tokens. Δ: FoveaWAM minus the strongest other method on that task; ties follow table order. The task-balanced row compares against the strongest overall baseline, PUMA. Brackets give pointwise 95% paired scene-bootstrap intervals, conditional on the selected comparator and checkpoint (100,000 resamples; no selection adjustment). The last two rows give first-contact position error e_p (mm), averaged over the 13 contact tasks, and the number n of episodes with a contact (Appendix B.3); –: no complete 13-task mean. Δ shading is blue for higher SR and orange for lower, saturating at 15 points. Blue/bold and light blue mark the best and second-best displayed values.

Task	ACT	DP	$\pi_{0.5}$	$\pi_{0.5}$+	DVLA	PUMA	OWAM	OW-off	DWAM	DWAM+	**Ours**	Δ [95% CI]
Harmonic grasp	49.0	31.0	51.0	84.0	39.0	85.0	41.0	1.0	50.0	47.0	**99.0**	+14.0 [7.0, 22.0]
Conveyor grasp	45.0	45.0	96.0	62.0	89.0	89.0	62.0	0.0	59.0	60.0	**100.0**	+4.0 [1.0, 8.0]
Translating grasp	14.0	33.0	59.0	42.0	61.0	51.0	53.0	0.0	53.0	18.0	**77.0**	+16.0 [4.0, 28.0]
Rotating grasp	24.0	20.0	32.0	53.0	26.0	67.0	42.0	3.0	30.0	35.0	**71.0**	+4.0 [-8.0, 16.0]
Occluded grasp	30.0	28.0	37.0	60.0	80.0	84.0	34.0	0.0	44.0	38.0	**100.0**	+16.0 [9.0, 23.0]
Moving receptacle	26.0	24.0	34.0	63.0	41.0	24.0	27.0	0.0	8.0	6.0	**93.0**	+30.0 [21.0, 40.0]
Rotating receptacle	0.0	0.0	1.0	11.0	1.0	0.0	0.0	0.0	0.0	0.0	**66.0**	+55.0 [45.0, 65.0]
Wire passage	99.0	99.0	**100.0**	89.0	99.0	87.0	50.0	58.0	23.0	78.0	**100.0**	+0.0 [0.0, 0.0]
Wire passage, full	0.0	3.0	1.0	0.0	12.0	0.0	0.0	0.0	0.0	0.0	**40.0**	+28.0 [16.0, 40.0]
Ball capture	36.0	16.0	**46.0**	29.0	41.0	39.0	45.0	13.0	24.0	20.0	38.0	-8.0 [-18.0, 2.0]
Ball capture, table	71.0	81.0	**100.0**	**100.0**	88.0	**100.0**	99.0	88.0	98.0	**100.0**	**100.0**	+0.0 [0.0, 0.0]
Slider capture	56.0	30.0	58.0	62.0	28.0	76.0	71.0	31.0	47.0	63.0	**79.0**	+3.0 [-5.0, 12.0]
Can capture	91.0	31.0	98.0	99.0	72.0	**100.0**	**100.0**	5.0	96.0	**100.0**	**100.0**	+0.0 [0.0, 0.0]
Post-impact grasp	42.0	5.0	49.0	68.0	39.0	57.0	21.0	22.0	43.0	19.0	**87.0**	+19.0 [7.0, 31.0]
Task-balanced	41.6	31.9	54.4	58.7	51.1	61.4	46.1	15.8	41.1	41.7	**82.1**	+20.8 [18.4, 23.1]
e_p (mm)	32.2	36.1	31.3	34.8	30.0	27.2	31.8	–	31.2	36.1	**22.0**	–
n	1110	946	1237	1191	1204	1272	1115	493	1193	1029	1285	–

flow control removes $\mathcal{L}_P$ and the pose head. It supervises flow at 4×4 points in each patch, forming a 56×56 grid with labels from the same ground-truth rigid motion. Table S7 reports the controls in all three settings, including paused physics.

Table S7. **Ablation controls in all three settings.** Task-balanced SR (%) of FoveaWAM on held-out OnTimeBench scenes and its change for each control (Δ, points). The Native, +150 ms, Rot., and Ctx. columns equal Table 3; Paused adds the paused-physics protocol. Red marks drops and green marks gains, with shading scaled within each column. Rot./Ctx.: rotation-sensitive and context-active tasks (Table S2).

	Success rate, all tasks			Groups (Native)	
	Paused	Native	+150 ms	Rot.	Ctx.
FoveaWAM	**90.3**	**82.1**	**34.1**	**80.4**	**82.7**
Precision allocation					
Uniform coarse flow	+4.4	−4.5	−0.5	−8.6	−4.6
Uniform fine flow	+1.3	+0.8	−0.9	−3.4	−2.6
w/o flow	+3.1	−4.2	−1.4	−11.7	−3.9
Target motion history					
w/o history	−19.5	−19.6	+0.2	−24.0	−2.3
Future supervision					
w/o future	+0.2	−4.3	−3.4	−12.6	−2.1
w/o $\mathcal{L}_{\text{aux}}$	−3.8	−2.6	−1.9	−11.6	+0.1

D. Real-Robot Experiments

D.1. Platform and Tasks

The robot is a Flexiv Rizon 4s arm with a GN01 gripper. Policies send absolute Cartesian tool-center-point poses and gripper targets at 20 Hz. Demonstration collection logs robot control at 100 Hz. The setup uses two RealSense D435 cameras, one fixed and one mounted on the wrist, both recording RGB at 640×480 and 60 Hz. We calibrate the fixed camera to the robot base with a ChArUco board. In the planned evaluation, FoveaWAM uses RGB-D frames and target poses from the fixed camera. Baselines use RGB from both cameras. The task devices are installed on this platform one at a time (Fig. 6; Table S8).

D.2. Pose Tracking and Labels

Offline labels. We label the recorded fixed view with masks from Grounding DINO [28] and SAM 2.1 [35]. FoundationPose [50] tracks the metric object mesh forward and backward using depth from FoundationStereo [51]. We check agreement between directions and use tool-center-point kinematics to fill occluded frames during rigid grasps. We retain 595 demonstrations with valid target poses on 97.1% of their frames. Tennis-ball labels use translation only. For sparse scene flow, we lift CoTracker3 [20] tracks

Figure placeholder: Real-robot failures

(a) Failures by category. One stacked bar per method. **(b) Representative trials.** Show unretouched fixed-view frames and mark the events that determine the outcomes.

Figure S1. **Real-robot failures.** (a) Failure counts by category and method. (b) Frames from representative trials. [Tasks, methods, and outcomes of the shown trials.]

Table S8. **Real-robot tasks.** A grasp succeeds when the robot holds the target stably and lifts it clear of its support. The time limit is 30 s for every task.

Task	Target motion	Success
Conveyor Grasp	Block on a conveyor belt	Grasp
Moving Box Placement	Open box on a conveyor belt	Block stays in box
Turntable Grasp	Rectangular block on a turntable	Grasp across short sides
Toy Car Grasp	Toy car moving across the table	Grasp
Tennis Ball Interception	Tennis ball rolls with randomized initial speed and heading offset	Stop or deflect with the gripper
Ramp Cart Grasp	Block carried by a cart on an incline	Grasp

to the robot base frame using stereo depth. Future slots are sampled at the 50–400 ms prediction horizons by interpolating valid observations.

Online pose estimation. We estimate target poses from the fixed camera's aligned RGB-D frames and transform them to the robot base frame using the camera-to-base calibration. The policy samples these estimates at 20 Hz to build its target-state history.

D.3. Training and Evaluation Protocol

The shared corpus contains 600 teleoperated demonstrations, with 101 for Turntable Grasp, 99 for Tennis Ball Interception, and 100 for each other task. ACT trains one policy per task for 50,000 updates. DynamicWAM fine-tunes one policy across the six tasks for 20,000 updates. Its deployment configuration uses asynchronous 16-action chunks, requests a new chunk with four commands (200 ms) remaining, and blends overlapping commands. Appendix A.2 describes fine-tuning for FoveaWAM.

We run 20 trials for each task and method in separate groups. An operator resets the scene before each trial and records its outcome. We count timeouts, operator interventions, and safety stops as failures. Table S6 reports robot timing.

D.4. Results and Failure Analysis

Table S9. **Real-robot success with intervals.** Success rates (%) pooled over the trials of each method, with 95% Wilson intervals. Table 4 gives per-task rates.

Method	All [95% CI]
$\pi_{0.5}$	—
DynamicWAM	—
ACT	—
FoveaWAM	—

Table S9 gives pooled success rates and intervals. Figure S1 groups failures by observable events, including late or misaligned grasps, slips or drops, collisions or safety stops, timeouts, and loss of target tracking. [The dominant failures by method and the evidence for a timing error.]